A Market-Based Analysis of Critical Nodes in Power Systems

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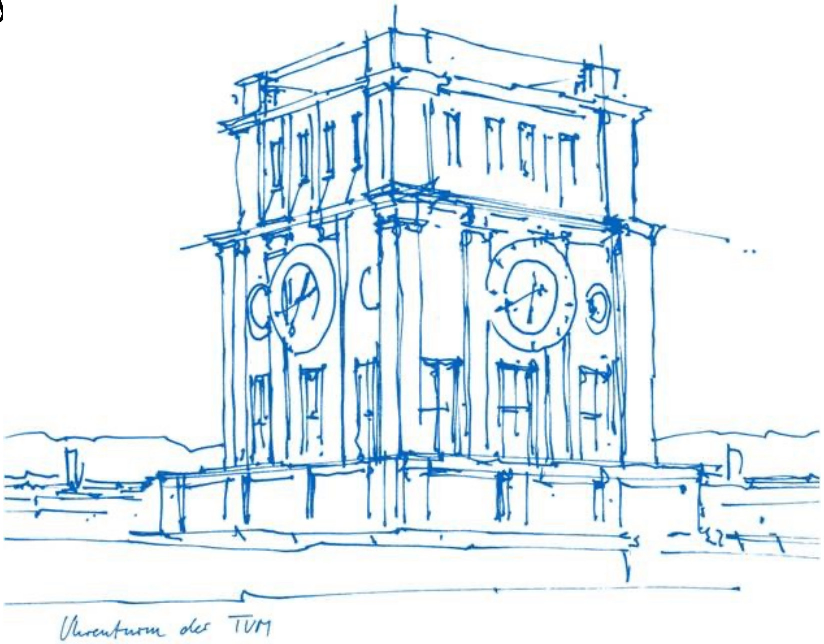


Motivation: Grid vulnerability and limited defensive resources

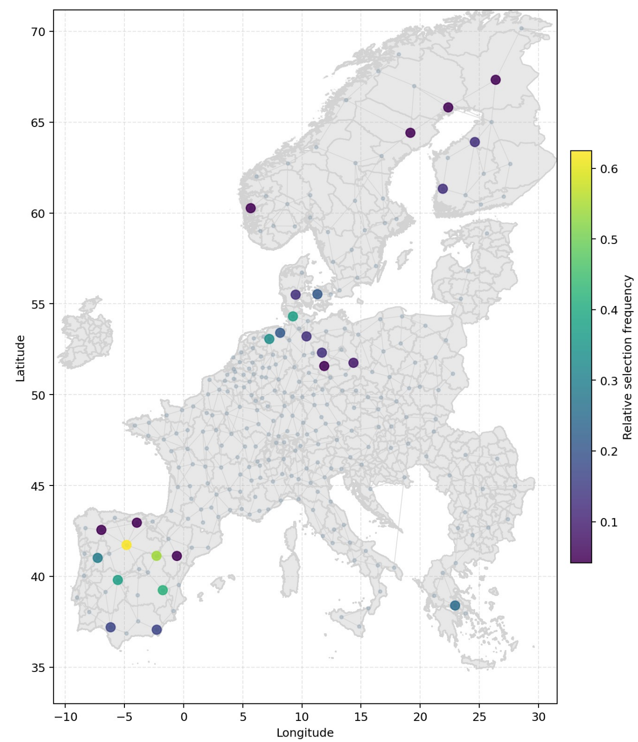
Critical infrastructure is increasingly exposed (renewables, digitization, cyber-physical complexity)

Real Attacks:

- Ukraine 2015 cyber-blackout[1]
- Berlin Jan 2026 high-voltage sabotage[2]

Economic damage can exceed $100M per event[3].

To allocate protective resources, the operator must identify the components that matter most.



Node relative selection frequency for all hours on 01.12.2013

[1]G. Liang, S. R. Weller, J. Zhao, F. Luo, and Z. Y. Dong. “The 2015 Ukraine Blackout: Implications for False Data Injection Attacks.” In: IEEE Transactions on Power Systems 32.4 (2017), pp. 3317–3318. doi: 10 . 1109 / TPWRS . 2016 .2631891.
[2]W. Rudischhauser, T. Fella, M. Łabuz, and M. Schulze. Bedingt resilient: Wasder Berliner Blackout Anfang 2026 für die deutsche Krisenvorsorge bedeutet. Arbeitspapier Sicherheitspolitik 1/2026. Bundesakademie für Sicherheitspolitik, Jan. 2026.
[3]Office of Technology Assessment. Physical Vulnerability of Electric Systems to Natural Disasters and Sabotage. Tech. rep. OTA-E-453. Washington, DC: U.S. Congress, Sept. 1990

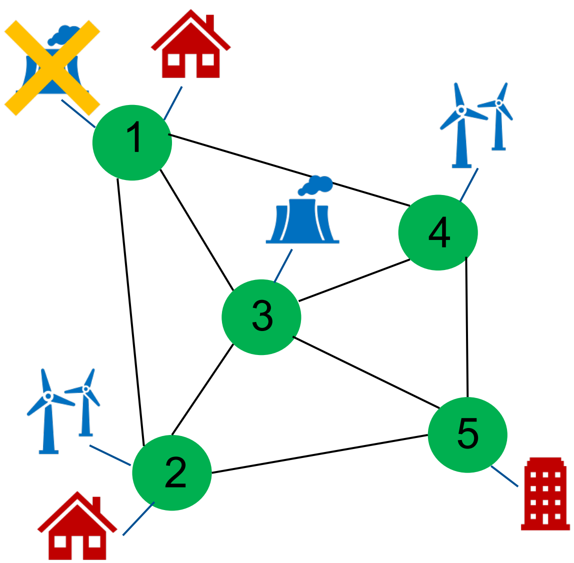
Component criticality: Two disruption types

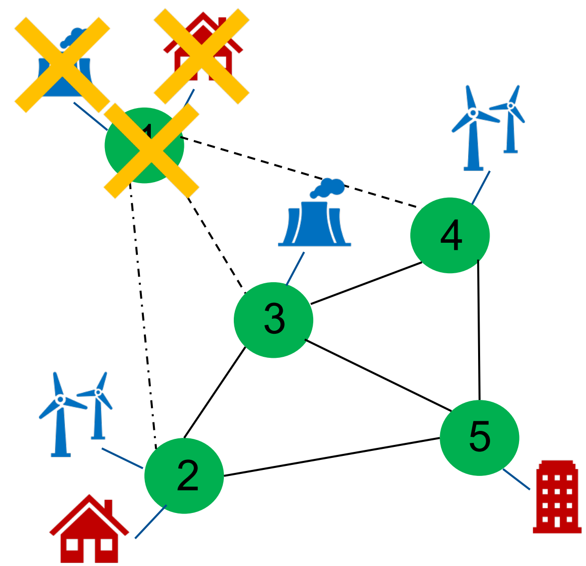
1. Generator-node interdiction (GNI)

- Remove generation capacity at selected nodes.
- Load and lines stay in the network.
- Effect: capacity shock.

2. Node Interdiction (NI)

- Remove node, generation, demand, and incident lines.
- Network topology changes.
- Effect: structural fragmentation.





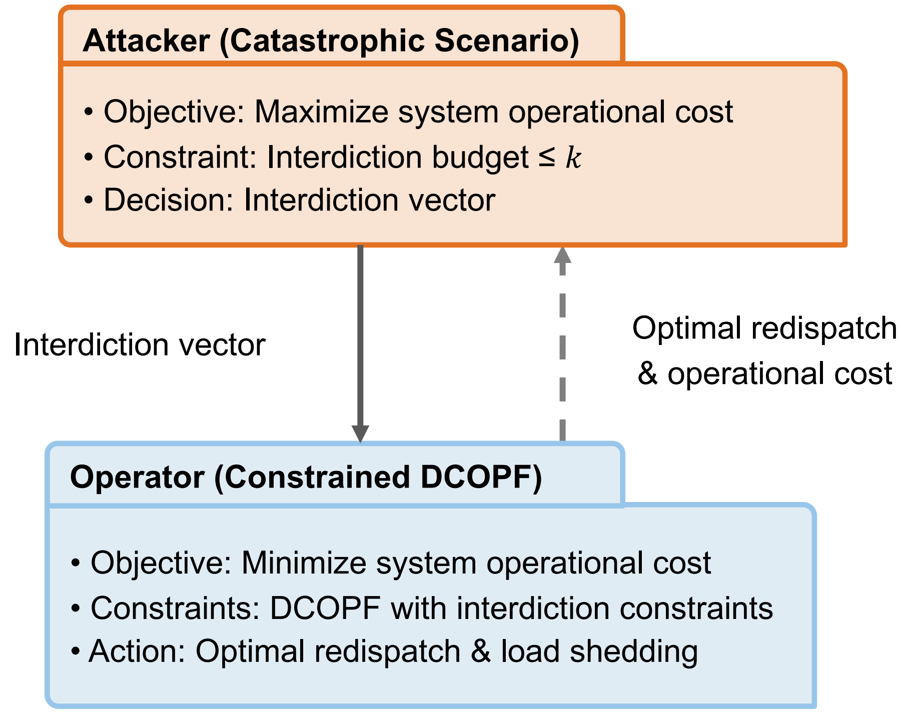
Bilevel Generator-Node Interdiction

Problem

- Attacker **disables *k* nodes**
- Defender re-dispatches via DCOPF to minimize cost
- Worst-case *k*-node combination = maximum economic damage

Solution Approach

- Bilevel problem can be reformulated as MILP: provably optimal, but expensive



Exact MILP interdiction is expensive for **realistic systems** and **large budgets *k*.**

Scalable screening heuristics

Problem: Exact MILP interdiction is expensive for **realistic systems** and **large budgets k.**

Solution: Use baseline information to **compute node criticality scores.**

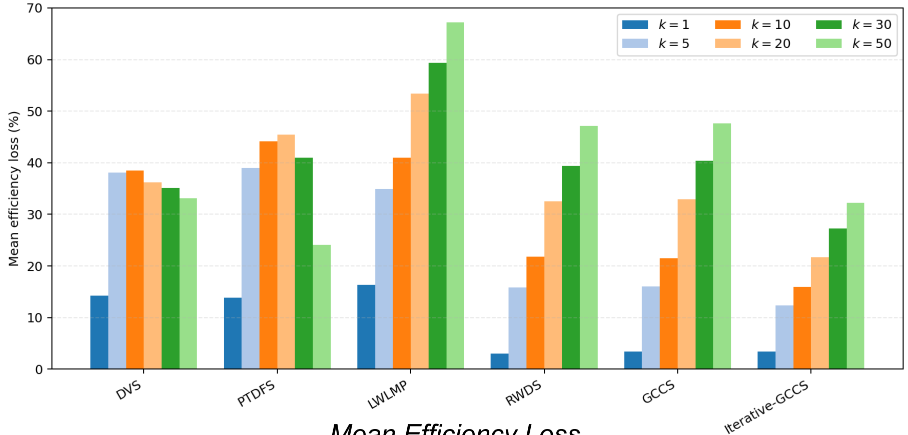
- Dispatch y_g^*
- Locational marginal price λ_v^*
- Generator capacity duals γ_g^*
- Congestion duals $\mu_{(v,w)}^*$
- PTDF

✓	$\sum_{g \in G(v)} y_g^*$	Dispatch Volume (DVS)
✓	$\sum_{g \in G(v)} \bar{P}_g \sum_l \frac{\lvert PTDF_{lv} \rvert}{m_l + \epsilon}$	PTDFS
✓	$\lambda_v^* D_v$	Load-Weighted LMP (LWMP)
✓	$\sum_{g \in G(v)} \left(\lambda_v^* - c_g\right) y_g^*$	Rent-Weighted Dispatch (RWDS)
✓	$\sum_{g \in G(v)} \gamma_g^* \bar{P}_g + \sum_{(v,w) \in \delta(v)} \bar{F}_{vw} \left(\mu_{(v,w)}^{+*} + \mu_{(v,w)}^{-*}\right)$	(GCCS)
✓	*Reoptimize after node interdiction*	Iterative-GCCS

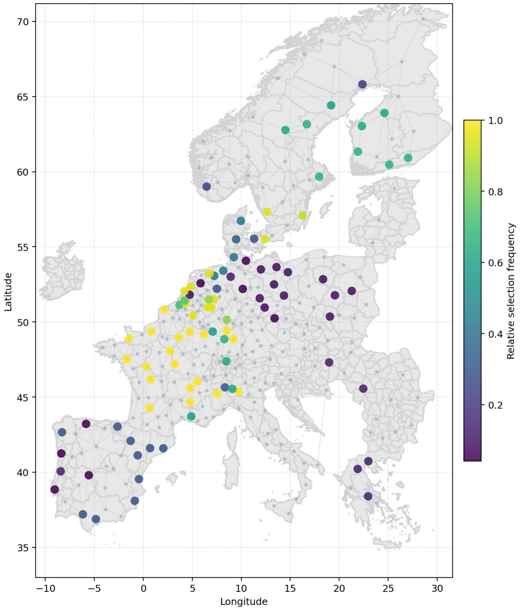
$m_l = \bar{F}_l - \lvert f_l^* \rvert$ line margin

$\bar{P}_g$ available capacity at gen. g

Generator-Node Interdiction Results



Mean Efficiency Loss

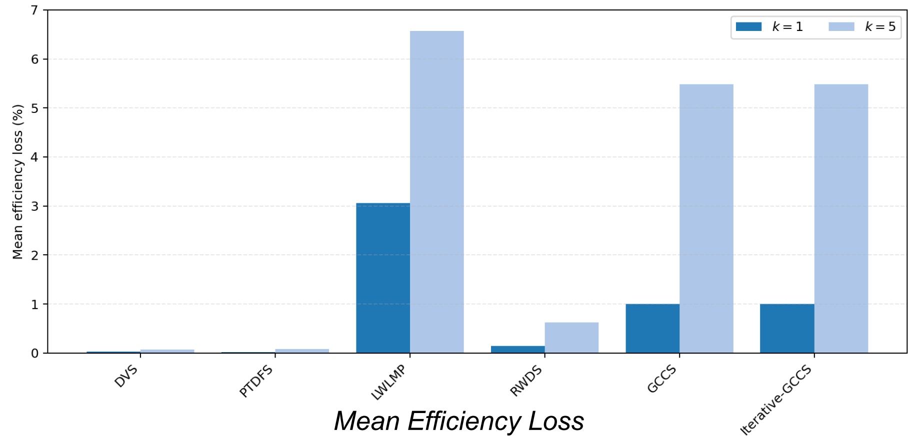


Node relative selection frequency for all hours on 01.12.2013

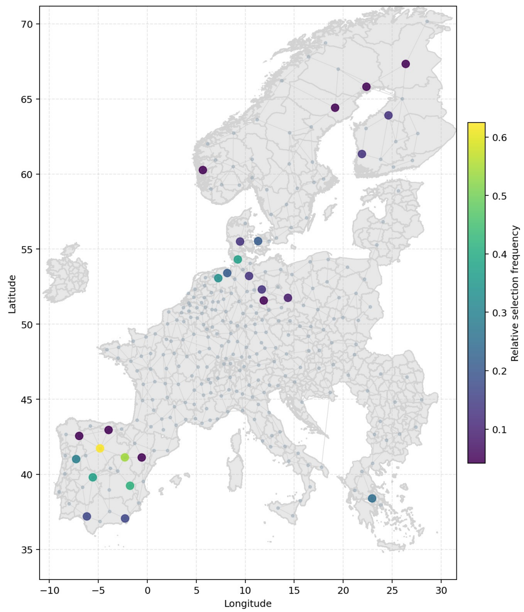
- Critical nodes are concentrated in generation-rich regions, but vary across operating conditions.
- Heuristics relying on dual variables provide accurate sensitivity information.
- Heuristics reduce runtime by up to 3–4 orders of magnitude compared to the exact MILP.

Heuristics were tested on a 300-node representation of the SDAC region, 240 periods (PyPSA-Eur).

Node Interdiction Results



Mean Efficiency Loss



Node relative selection frequency for all hours on 01.12.2013

- Critical nodes are more spatially dispersed and primarily reflect connectivity-driven rather than scarcity-driven vulnerability.

- Topology changes make baseline dual information unreliable; flow-based methods (especially PTDFS) outperform economic heuristics.

Heuristics were tested on a 300-node representation of the SDAC region, 240 periods (PyPSA-Eur).

Conclusions

- Different disruption mechanisms require different resilience metrics.
- Dual-based heuristics work well for generator-node interdictions.
- Flow-based metrics work better for node interdictions.
- LP-based screening methods scale well to large systems.
- Resilience planning should distinguish between scarcity-driven and connectivity-driven disruptions.

Thank you!